

Linguistic Holonomy and the Erosion of Statistical Watermarks: Meaning-Preserving Transformations as Paths, not Endpoints

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Abstract

Statistical watermarks for language models live in the freedom of the signifier: they choose among tokens that are nearly equivalent in meaning, and they are therefore eroded by exactly those transformations which move the form of a text while leaving its content in place. The literature measures such transformations by their endpoint, through the semantic similarity between the original and the rewritten text. We show that the endpoint is the wrong statistic. Adapting the formalism of linguistic loops, we prove that the invariant of a chain of meaning-preserving transformations factorises canonically into an endpoint part and a holonomy in the stabiliser of the initial state, the second of which the semantic deficit cannot see; the loop rotation is parallel transport on the unit sphere of the embedding space, so that the analogy with the Wilson loop becomes a theorem rather than a figure of speech. On the side of the detector we prove an exact identity: the residual statistic is proportional to the number of positions whose seeding window survived intact, from which the decay law ρ^{h+1} follows as the independent-edit corollary. The identity has a disconcerting consequence, which we confirm to three decimal places: at one and the same retention rate the surviving signal may be one half of the original, one quarter of it, or exactly nothing, according only to where the edits fall.

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1 Introduction and Motivation

When Raymond Queneau published his *Exercices de style* in 1947, inspired by Bach's *Art of Fugue*, he wrote the same trivial anecdote ninety-nine times. A man is jostled on a bus; two hours later a friend advises him to move a button on his overcoat. Nothing else happens. What varies is everything else: the register, the tense, the person, the alphabet, the sonnet form, the language of a mathematician. The content is held fixed

Family	Mechanism	Context h	Output law	What an edit destroys
Green list [7]	logit bias	≥ 1	perturbed	token <i>and</i> its h predecessors
Unigram [15]	logit bias	0	perturbed	the token alone
Exponential [10, 2]	sampling rule	≥ 0	exact	token <i>and</i> seeding context
Tournament [5]	sampling rule	≥ 1	exact	token <i>and</i> seeding context
Semantic [6]	sentence partition	—	perturbed	the sentence embedding

Table 1: *Synoptic table of the families of text watermarks. The last column is the one that matters for this article: in every family except the Unigram scheme, an edit damages not only the position it touches but every position that used it as seeding context. It is this asymmetry which the geometry of Section 4 and the identity of Section 5 make precise.*

by fiat, and the form is allowed to move as far as it can. Queneau took himself to be doing something intrinsically geometric, and named an early version of the work *Dodécaèdre*; the intuition, as was argued in [4], is not casual at all.

Eighty years later the same operation has acquired an adversarial use. Since 2 August 2026 the outputs of at least one major language model carry a machine-readable mark, in response to the transparency obligations of Article 50 of the European AI Act [1]. The published description of such marks is careful: the mark does not change meaning, quality or readability, it survives copying and light editing, and it is stated to be lost under paraphrase, heavy editing, translation and mixing with other text. Whoever wishes to remove the mark, then, is being invited to perform an *Exercice de style*. The question of how much of the mark survives such an exercise is the subject of this article.

The landscape

The dominant construction is the *green-list* watermark of Kirchenbauer *et al.* [7]: at each step a pseudo-random subset of the vocabulary, of relative size γ , is declared green on the basis of a secret key and of the h preceding tokens, and the logits of green tokens are raised by a bias δ_b . Detection counts green tokens and reads off a z -score. Setting $h = 0$ gives the Unigram scheme of Zhao *et al.* [15], whose robustness is provable. A different family, initiated by Aaronson and developed by Kuditipudi *et al.* [10] and by Christ, Gunn and Zamir [2], leaves the output distribution mathematically untouched and replaces instead the source of randomness in the sampler; these are the *distortion-free* schemes. A production deployment at scale, using tournament sampling, is described in [5]. A synoptic view is given in Table 1.

Against these constructions stands a literature of attacks. Krishna *et al.* [9] introduced a dedicated paraphraser; Sadasivan *et al.* [13] iterated it; a recent large-scale study [3] chains rewritings up to five deep and reports detection falling from 87.9% on the original outputs to 4.86% after five hops. Kirchenbauer *et al.* [8] had already observed, on empirical grounds, that a longer seeding context makes a watermark less robust to editing.

All of these studies share a habit of measurement. An attack is scored by its *endpoint*: by the semantic similarity between the original and the final text, and by a count of how many tokens or hops separate them. The intermediate states are treated as scaffolding.

The gap

This habit is not innocent. Consider two chains of rewritings that begin at the same text and end at texts of identical meaning. One proceeds directly; the other wanders through three languages and returns. By any endpoint measure they are the same attack. Are they?

To ask the question one needs a formalism in which the intermediate states are part of the object rather than part of the apparatus. Such a formalism exists. In [4] a *linguistic loop* was defined as a chain of transformations that move the signifier while preserving the semantic core, its *semantic deficit* was defined as the cosine distance between the first and last embedded states, and a rotation $R_{\mathcal{U}} \in \text{SO}(n)$ was associated with the entire chain by composing the minimal rotations between consecutive states. The signature of a quadratic form built from $R_{\mathcal{U}}$ was proposed as the invariant of the loop, and the construction was compared, in the introduction of that paper, with the Wilson loop of lattice gauge theory.

We take up that formalism, and we find three things.

The first is that the proposed invariant collapses. The representing matrix $I_n - R_{\mathcal{U}}^*$ is always positive semidefinite, so its Sylvester signature is $(2p, 0, n - 2p)$ and carries no more information than its rank. The negative index, which is where a signature normally keeps its content, is identically zero.

The second is that the analogy with the Wilson loop is not an analogy. The minimal rotation of [4] is parallel transport along the minimising geodesic of the unit sphere; $R_{\mathcal{U}}$ is parallel transport along the geodesic polygon of the chain; and the quantity that the semantic deficit discards is precisely a holonomy, an element of $\text{SO}(n - 1)$ fixing the initial state. Endpoint and path separate canonically, and they are independent: neither constrains the other.

The third is that on the side of the detector there is an exact law, and it is not the law the field has been assuming. The residual statistic after an attack is proportional to the number of scored positions whose entire seeding window survived. When the edits are independent this gives ρ^{h+1} , which explains in closed form the trend reported in [8]. When they are not — and real edits never are — the retention rate ρ does not determine the residual at all.

Contribution

In this work we prove that the invariant of a linguistic loop factorises canonically as $R_{\mathcal{U}} = R_{\text{dir}} H$ with H in the stabiliser of the base point, that the semantic deficit depends only on the first factor, and that the two factors are functionally independent; we identify H as the Riemannian holonomy of the geodesic polygon of the chain, which on the two-sphere is the area of the enclosed triangle; we show that the signature invariant of [4] degenerates to a rank and is superseded by the angle spectrum; and we prove the intact-window identity for green-list and exponential detectors, with its corollary that an adversary who knows the context width h can annihilate the signal while retaining a fraction $1 - 1/(h + 1)$ of the tokens. Every statement is verified numerically against exact detectors whose keys we hold, and the geometric machinery is applied to real chains of round-trip machine translation.

In Section 2 we recall, in a self-contained form, the apparatus of linguistic loops. In Section 3 we prove the degeneracy of the signature. Section 4 contains the factorisation, the identification with parallel transport and the Gauss–Bonnet corollary. Section 5

turns to the detector and proves the intact-window law. Section 6 reports the experiments, including a preregistered test whose outcome we report whichever way it fell. Section 7 discusses what the results do and do not license.

2 Linguistic loops and their invariants

We recall the apparatus of [4], in the form in which we shall use it. The reader who knows that paper may skip to Section 3; the reader who does not should find here everything the sequel requires.

Language is not a metric space, but its images under an embedding map are. A *metrizable linguistic space* is a triple $(\mathcal{A}, \psi, d)$ where $\mathcal{A}$ is a set of linguistic elements — words, phrases, whole propositions — in a context of interest, and

$$\psi : \mathcal{A} \longrightarrow \mathbb{R}^n, \quad \lambda \longmapsto \psi(\lambda), \quad (2.1)$$

is an embedding map, generally neither injective nor surjective. A distance d_* on $\mathbb{R}^n$ induces a *semantic distance* on $\mathcal{A}$ by composition,

$$d(\lambda, \nu) := d_*(\psi(\lambda), \psi(\nu)), \quad (2.2)$$

and throughout this article d_* is the cosine distance

$$d_{*,\cos}(x, y) = 1 - \frac{x \cdot y}{\|x\| \|y\|}. \quad (2.3)$$

The choice is not innocuous and we shall return to it in Remark 3.2: it is what makes the whole construction orthogonally natural rather than linearly natural.

A *linguistic transformation* U is a map of $\mathcal{A}$ into itself which acts on the signifier while preserving, in a controlled way, the semantic core. In [4] three requirements are imposed: closure, $U(\lambda) \in \mathcal{A}$; reversibility, the existence of an approximate inverse U^{-1} with $d(U^{-1}(U(\lambda)), \lambda) < \varepsilon$ for a fixed threshold ε ; and coherence, the requirement that similar elements have similar images. Translation between languages, conversion to the negative or interrogative form, dialectal or stylistic adaptation, expansion and synthesis, and paraphrase are all linguistic transformations; inventing a story is not, because its generative content is too large for the original to be reconstructible.

Given a sequence $\mathcal{U} = \{\mathcal{K}, U_1, \dots, U_L\}$ of such transformations and an initial element λ , the iterated composition produces a chain in $\mathcal{A}$, whose image under ψ is a chain of vectors

$$\psi(\mathcal{U}(\lambda)) = \{v_0, v_1, \dots, v_L\}, \quad v_0 = \psi(\lambda), \quad v_i = \psi((U_i \circ \dots \circ U_1)(\lambda)). \quad (2.4)$$

The *semantic deficit* of the chain is the distance between its extremes,

$$\delta_{\mathcal{U}}(\lambda) := d_*(v_0, v_L), \quad (2.5)$$

and when $\delta_{\mathcal{U}} < \xi$ for a fixed threshold ξ the sequence is called a *linguistic loop*. Small deficit means the meaning came back; the form, in the meantime, may have gone anywhere.

To capture where it went, one rewrites the chain as a sequence of rotations. For $x, y \in \mathbb{R}^n$ neither zero nor antipodal, write $\hat{x} = x/\|x\|$ and let

$$\tilde{\mathcal{R}}^{x,y} := I_n + (\hat{y}\hat{x}^T - \hat{x}\hat{y}^T) + \frac{1}{1 + \hat{x}^T \hat{y}} (\hat{y}\hat{x}^T - \hat{x}\hat{y}^T)^2 \in \text{SO}(n) \quad (2.6)$$

be the *minimal rotation* carrying $\hat{x}$ to $\hat{y}$. Its minimality is the statement that it rotates in the plane spanned by x and y only, acting as the identity on the orthogonal complement. The whole chain is then summarised by the ordered product

$$R_{\mathcal{U}} := \tilde{\mathcal{R}}^{v_{L-1}, v_L} \tilde{\mathcal{R}}^{v_{L-2}, v_{L-1}} \dots \tilde{\mathcal{R}}^{v_0, v_1} \in \text{SO}(n), \quad (2.7)$$

which satisfies $R_{\mathcal{U}} \hat{v}_0 = \hat{v}_L$ and, unlike the deficit, remembers the intermediate states. Writing $R_{\mathcal{U}}^* = \frac{1}{2}(R_{\mathcal{U}} + R_{\mathcal{U}}^T)$ for the symmetric part, an elementary computation recovers the deficit as a quadratic form,

$$\delta_{\mathcal{U}}(\lambda) = Q_{\mathcal{U}}(\hat{v}_0, \hat{v}_0), \quad Q_{\mathcal{U}}(\hat{v}, \hat{v}) := \hat{v}^T (I_n - R_{\mathcal{U}}^*) \hat{v}, \quad (2.8)$$

whose representing matrix is the real symmetric $I_n - R_{\mathcal{U}}^*$. It was proposed in [4] that the Sylvester signature of this matrix be taken as the invariant of the loop, on the ground that it probes finer structural properties than the deficit alone. That it probes finer properties is true. How much finer is the subject of the next section.

3 The signature and its degeneracy

A signature keeps its information in the interplay between its positive and negative indices; Sylvester's law of inertia is informative because a form can be indefinite. The forms which arise from linguistic loops, however, cannot.

Proposition 3.1 (Degeneracy of the signature). *Let $R \in \text{SO}(n)$ with non-trivial rotation angles $\theta_1, \dots, \theta_p \in (0, \pi]$, and set $M = I_n - \frac{1}{2}(R + R^T)$. Then M is positive semidefinite, its spectrum is*

$$\{1 - \cos \theta_k \text{ with multiplicity } 2\}_{k=1}^p \cup \{0 \text{ with multiplicity } n - 2p\}, \quad (3.1)$$

and consequently

$$\text{sign}(M) = (2p, 0, n - 2p). \quad (3.2)$$

The negative index vanishes identically, $\text{rank}(M) = 2p$, and the signature is a function of the single integer p .

Proof. By the real normal form of a special orthogonal matrix there is $Q \in \text{O}(n)$ with $Q^T R Q = \text{diag}(R(\theta_1), \dots, R(\theta_p), I_{n-2p})$, where $R(\theta)$ is the planar rotation by θ . Angles are taken in $(0, \pi]$: the eigenvalue -1 of an element of $\text{SO}(n)$ occurs with even multiplicity, and each pair constitutes a block $R(\pi)$, while $\theta = 0$ contributes a trivial block absorbed into I_{n-2p} . Since $R(\theta)^T = R(-\theta)$, the symmetric part of a block is $\cos(\theta)I_2$, whence

$$Q^T M Q = \text{diag}((1 - \cos \theta_1)I_2, \dots, (1 - \cos \theta_p)I_2, 0). \quad (3.3)$$

As Q is orthogonal, (3.3) is at once a congruence and a similarity, so its diagonal entries are the eigenvalues of M , which is (3.1); and $1 - \cos \theta_k > 0$ for $\theta_k \in (0, \pi]$, which gives positive semidefiniteness and (3.2). $\square$

The proof is four lines, and this is itself the point: nothing in the construction could have produced an indefinite form, because a rotation never moves a vector further than antipodally. It is worth stating what survives. The integer p , the number of two-planes in which the chain has genuinely rotated, is a real invariant of the loop, and it is not

visible in the deficit; what does not survive is the expectation that a signature carries more than a rank.

The natural replacement is not far to seek. Since $Q^T R Q$ is determined up to permutation of the blocks by the multiset of angles, the complete invariant of R under conjugation is the *angle spectrum*

$$\Theta(R) := \{\theta_1, \dots, \theta_p\}, \quad (3.4)$$

equivalently the conjugacy class of R in $\text{SO}(n)$, of which (3.2) retains only the cardinality. That the refinement is strict is immediate: for $n \geq 4$ any two rotations with two non-trivial planes and different angles share the signature $(4, 0, n - 4)$ and differ in Θ . We shall use throughout the scalar reduction

$$\|\Theta(R)\|_2 = \left(\sum_{k=1}^p \theta_k^2 \right)^{1/2}, \quad (3.5)$$

which we call the *rotation energy* of R , with the warning that it is a lossy summary of Θ and is used for convenience of regression, not because it is canonical.

Remark 3.2 (The invariance group). It is of paramount importance to notice which group is acting. The signature of a *fixed* quadratic form is invariant under congruence $M \mapsto P^T M P$ for any $P \in \text{GL}(n, \mathbb{R})$; this is Sylvester's law and it is not in question. But the assignment which sends a chain to its form is not $\text{GL}(n, \mathbb{R})$ -natural. If $A \in \text{O}(n)$ then $\widehat{A}x = A\hat{x}$ and $\hat{x} \cdot \hat{y}$ is preserved, so (2.6) transforms by conjugation, $\tilde{\mathcal{R}}^{Ax, Ay} = A\tilde{\mathcal{R}}^{x, y}A^T$, and therefore so does (2.7); the angle spectrum is an invariant of the chain. For a general $A \in \text{GL}(n, \mathbb{R})$ neither identity holds, because (2.3) is not preserved, and the loop matrix of the transformed chain is unrelated to the original. The natural invariance group of the construction is thus $\text{O}(n)$, and the invariant it protects is Θ , not the signature.

Remark 3.3 (A trap in the numerics). The identity $n_+ = 2p$ holds numerically only if the two counts are taken with matched tolerances. Declaring a plane non-trivial when $\theta > \text{tol}$, while declaring an eigenvalue positive when $1 - \cos \theta > \text{tol}$, admits to the first count planes with $\text{tol} < \theta < \sqrt{2}\text{tol}$ which the second legitimately rejects; the angle threshold must be $\arccos(1 - \text{tol})$. Moreover a plane rotated by exactly π has both its eigenvalues equal to -1 , so a count based on the arguments of the eigenvalues reports the angle π twice for a single plane. We record this because both traps were sprung in the course of the verification reported in Section 6.

4 The holonomy of a linguistic loop

The deficit (2.5) is a function of the two extreme states. The loop matrix (2.7) is a function of all of them. The difference between the two is an object, and it is a familiar one.

We begin by observing that the minimal rotation was not an arbitrary choice of interpolant.

Lemma 4.1 (The minimal rotation is parallel transport). *Let $\hat{x}, \hat{y}$ be distinct, non-antipodal unit vectors of $\mathbb{R}^n$ and $P = \text{span}(\hat{x}, \hat{y})$. Then $\tilde{\mathcal{R}}^{x, y}$ is the parallel transport of the round metric of S^{n-1} along the minimising geodesic from $\hat{x}$ to $\hat{y}$, extended to $\mathbb{R}^n$ as the ambient rotation fixing $P^\perp$ pointwise.*

Proof. The minimising geodesic is the arc of the great circle $S^{n-1} \cap P$. A field W along a geodesic c of the round sphere is parallel exactly when its ambient derivative is normal to the sphere, that is when $W' = -\langle W, c' \rangle c$. Decomposing $W = W_P + W_\perp$, the component $W_\perp$ is constant, since c and c' lie in P , while W_P rotates inside P by the arclength travelled. Transport is therefore the rotation by $\theta = \arccos(\hat{x} \cdot \hat{y})$ inside P together with the identity on $P^\perp$. Writing $A = \hat{y}\hat{x}^T - \hat{x}\hat{y}^T$ one has $AP^\perp = 0$, so (2.6) is the identity on $P^\perp$; on P it carries $\hat{x}$ to $\hat{y}$ preserving orientation and metric, hence it is the rotation by θ . $\square$

Thus $R_{\mathcal{U}}$ is parallel transport along the geodesic polygon joining $\hat{v}_0, \hat{v}_1, \dots, \hat{v}_L$ on the unit sphere of the embedding space, and the chain of reformulations is literally a path on a sphere. The Wilson loop of [4] was not a metaphor.

Proposition 4.2 (Endpoint–path factorisation). *Let $V = (v_0, \dots, v_L)$ be a chain as in (2.4), with $n \geq 3$ and no two consecutive states antipodal. Put $R_{\text{dir}} = \tilde{\mathcal{R}}^{v_0, v_L}$ and*

$$H := R_{\text{dir}}^{-1} R_{\mathcal{U}}. \quad (4.1)$$

Then

- (i) $H\hat{v}_0 = \hat{v}_0$, so that $H \in \text{Stab}(\hat{v}_0) \cong \text{SO}(n-1)$, and $R_{\mathcal{U}} = R_{\text{dir}}H$ canonically;
- (ii) $\delta_{\mathcal{U}} = 1 - \langle \hat{v}_0, R_{\text{dir}}\hat{v}_0 \rangle$ depends only on R_{dir} , and is therefore blind to H ;
- (iii) H is the Riemannian holonomy, based at $\hat{v}_0$, of the closed geodesic polygon obtained by closing the path with the geodesic from $\hat{v}_L$ back to $\hat{v}_0$;
- (iv) for every $u \in S^{n-1}$ and every $H_0 \in \text{Stab}(\hat{v}_0)$ there is a chain with initial state $\hat{v}_0$, final state u and holonomy exactly H_0 .

Proof. (i) By construction $\tilde{\mathcal{R}}^{v_{i-1}, v_i} \hat{v}_{i-1} = \hat{v}_i$, so by induction $R_{\mathcal{U}}\hat{v}_0 = \hat{v}_L$; and $R_{\text{dir}}\hat{v}_0 = \hat{v}_L$ by definition. Since R_{dir} is orthogonal, $H\hat{v}_0 = R_{\text{dir}}^T \hat{v}_L = \hat{v}_0$. An element of $\text{SO}(n)$ fixing a unit vector preserves its orthogonal hyperplane and restricts there to an element of $\text{SO}(n-1)$.

(ii) Immediate from (2.5), (2.3) and $\hat{v}_L = R_{\text{dir}}\hat{v}_0$.

(iii) By Lemma 4.1, $R_{\mathcal{U}}$ and R_{dir} are the transports along the two paths, and (4.1) is the transport around the closed circuit.

(iv) It suffices to realise every H_0 as the holonomy of a closed geodesic polygon based at $\hat{v}_0$: appending the single geodesic leg from $\hat{v}_0$ to u multiplies the loop matrix on the left by $\tilde{\mathcal{R}}^{\hat{v}_0, u}$ and leaves the holonomy unchanged. Fix a two-plane $Q \subset \hat{v}_0^\perp$ and an angle $\alpha \in (0, 2\pi)$, and let $\Sigma = S^{n-1} \cap (\text{span}(\hat{v}_0) \oplus Q)$, a totally geodesic two-sphere through $\hat{v}_0$ whose tangent space there is Q . A geodesic triangle in Σ with vertex $\hat{v}_0$ and area α exists, since the area of a geodesic triangle on the unit two-sphere sweeps the whole of $(0, 2\pi)$; transport around a loop contained in a totally geodesic submanifold is the transport computed inside it, extended by the identity on the normal directions, so by Corollary 4.3 below the holonomy is the rotation of Q by α and the identity elsewhere. By the normal form used in Proposition 3.1, every element of $\text{SO}(n-1)$ is a product of at most $\lfloor (n-1)/2 \rfloor$ such plane rotations, and the corresponding polygons, each beginning and ending at $\hat{v}_0$, may be concatenated. $\square$

Part (iv) deserves a word, because it is what makes the factorisation worth having. It says that the endpoint datum and the path datum are *free*: prescribing how far the

meaning has drifted places no constraint whatever on how far the form has wandered, and conversely. Incidentally the proof re-establishes, without appeal to the classification of symmetric spaces, that $\text{Hol}(S^{n-1}, \hat{v}_0) = \text{SO}(n-1)$.

Corollary 4.3 (Gauss–Bonnet). *For $n = 3$ and $L = 2$ the holonomy is the rotation of the tangent plane at $\hat{v}_0$ by the spherical excess of the geodesic triangle $\hat{v}_0\hat{v}_1\hat{v}_2$, that is by its area.*

Proof. Gauss–Bonnet with Gaussian curvature 1, the excess of a geodesic triangle being $A + B + C - \pi$. $\square$

This is the statement to keep in mind, and it is worth dwelling on. On the two-sphere the invariant that the semantic deficit throws away is the *area swept* by the chain of reformulations. Two paraphrase chains ending at the same meaning differ by the area they enclose. One should add, since the numerics will otherwise appear to fail, that the angle of a rotation is recovered from its eigenvalues only as a principal value in $(0, \pi]$, so the holonomy determines the area outright when the triangle covers at most a hemisphere and modulo 2π in general.

We shall use the scalar

$$\eta(\mathcal{U}) := \|\Theta(H)\|_2 \quad (4.2)$$

and call it the *holonomy energy* of the chain. It is, again, a lossy reduction: a null result for η is not a null result for H .

5 Watermarks as functionals on the loop

We now cross from the semantic channel to the channel in which a watermark actually lives. The two are complementary, and the complementarity is the conceptual heart of this article: what a meaning-preserving chain *preserves* is the semantic core, and what a watermark *occupies* is precisely the freedom that remains once the meaning is fixed. The invariant and the carrier are, so to speak, dual coordinates on the same transformation. A loop with small deficit and large holonomy is, from the point of view of the detector, the worst case.

5.1 The intact-window law

Fix a green-list scheme with green fraction γ , context width $h \geq 0$ and bias δ_b , detecting a sequence $y_0, \dots, y_{T-1}$ by

$$z = \frac{G - \gamma T'}{\sqrt{T'\gamma(1-\gamma)}}, \quad T' := T - h, \quad (5.1)$$

where G counts the scored positions $t = h, \dots, T-1$ at which y_t lies in the green list seeded by $(y_{t-h}, \dots, y_{t-1})$. Let an attack replace the tokens at a set E of positions without changing the length, and define the *intact-window set*

$$I := \{t : h \leq t < T, [t-h, t] \cap E = \emptyset\}. \quad (5.2)$$

We assume, as is standard, that the hash behaves as a random oracle, so that green membership is an independent Bernoulli(γ) across distinct pairs of seed and token; and that at each scored position of unattacked watermarked text the green indicator is Bernoulli(γ_w) with $\gamma_w > \gamma$, independently across positions. The second is a mean-field hypothesis and is the weaker of the two; we return to it below.

Theorem 5.1 (Intact-window law). *Under the two hypotheses above,*

$$\mathbb{E}[z_{\text{att}}] = \frac{|I|}{T'} \mathbb{E}[z_0]. \quad (5.3)$$

Proof. Fix a scored position t . If $t \in I$ then y_t and its whole seeding window are those the generator produced, so the pair evaluated by the detector is the pair the generator evaluated, and it is green with probability γ_w . If $t \notin I$ then either the window or the token differs, so the pair is one the generator never biased, and by the random-oracle hypothesis its green indicator is a fresh Bernoulli(γ). Summing,

$$\mathbb{E}[G_{\text{att}}] = |I|\gamma_w + (T' - |I|)\gamma = \gamma T' + |I|(\gamma_w - \gamma), \quad (5.4)$$

and substituting into (5.1) gives $\mathbb{E}[z_{\text{att}}] = |I|(\gamma_w - \gamma)/\sqrt{T'\gamma(1 - \gamma)}$. The same computation with $|I| = T'$ gives $\mathbb{E}[z_0]$, and the ratio is (5.3). $\square$

The identity is deterministic in $|I|$; no distribution over the edits has been assumed. It is only when one wishes to compute $|I|$ that a model of the attack becomes necessary, and the model the field has implicitly been using is the independent one.

Corollary 5.2 (Context-width decay law). *If each position is retained independently with probability ρ , then $\mathbb{E}|I| = \rho^{h+1}T'$ and*

$$\mathbb{E}[z_{\text{att}}] = \rho^{h+1} \mathbb{E}[z_0]. \quad (5.5)$$

The signal decays exponentially in the context width and only linearly in the retention rate.

Equation (5.5) is, in closed form, the trend that [8] reported empirically. It also explains a phenomenon familiar to anyone who has attacked such a scheme by hand: the signal falls considerably faster than the fraction of altered words, and it falls faster the longer the seeding context. Since $\mathbb{E}[z_0]$ grows as $\sqrt{T'}$, one obtains at once the length required for detection at a fixed threshold z^* ,

$$T' \geq \frac{(z^*)^2 \gamma (1 - \gamma)}{(\rho^{h+1}(\gamma_w - \gamma))^2}, \quad (5.6)$$

which grows like $\rho^{-2(h+1)}$ and quantifies the published warning that short passages carry no reliable signal.

5.2 Why the retention rate is the wrong statistic

Real edits are not independent. A translator rewrites clauses, not tokens; a human editor works on paragraphs. The following is therefore not a curiosity but the typical case.

Proposition 5.3 (Arrangement dominates the rate). *Fix the retention rate $\rho = 1 - |E|/T$. Then*

- (i) *if E is a single contiguous run, $|I| \geq T' - |E| - h$, so the residual ratio is at least $\rho - h/T'$, independently of h ;*
- (ii) *if E consists of b maximal contiguous runs, $|I| \geq T' - |E| - hb$;*

(iii) if E contains an arithmetic progression of step $k \leq h + 1$ covering $[0, T)$, then $I = \emptyset$ and the expected residual is exactly zero.

Consequently, whenever $1 - \rho \geq 1/(h + 1)$, the residual ratio ranges over essentially the whole of $[0, \rho]$ as the arrangement varies at fixed ρ .

Proof. A scored position leaves I only if its window of $h+1$ consecutive positions meets E . For a single run of length $|E|$ the windows meeting it are those with $t \in [\min E, \max E + h]$, at most $|E| + h$ of them, which is (i); with b runs the bound is additive, which is (ii). If E contains every k -th position with $k \leq h + 1$, every window of $h + 1$ consecutive positions contains a member of E , so $I = \emptyset$ and Theorem 5.1 gives zero. $\square$

The consequence is worth stating without euphemism. An adversary who knows h — and h is a published design parameter, not a secret — can reduce the expected detector statistic to zero while retaining a fraction $1 - 1/(h + 1)$ of the tokens. For the common choice $h = 1$ this means editing one token in two; for $h = 3$, one in four. Nothing in the attack requires knowledge of the key. This is not a new attack so much as an exact accounting of a known design tension [8, 11], but we have not found it written down in this form, and its corollary — that a robustness table indexed by retention rate or by endpoint similarity is under-specified — appears not to have been drawn.

Remark 5.4 (The exponential family). Nothing in the proof of Theorem 5.1 used the biasing mechanism; only the context seeding was used. For the distortion-free scheme which seeds a uniform vector ξ from the context and scores $S = \sum_t -\log(1 - \xi_{y_t})$, whose null mean is T' , the same conditioning gives

$$\mathbb{E}[S_{\text{att}}] - T' = \frac{|I|}{T'} (\mathbb{E}[S_0] - T'), \quad (5.7)$$

and with it the analogues of Corollary 5.2 and Proposition 5.3. Leaving the output distribution mathematically untouched buys nothing at all against this particular weakness: the vulnerability is a property of context seeding, not of distortion.

5.3 Scope

It is worth being explicit about what has *not* been proved. Theorem 5.1 concerns substitutions that preserve length. Insertions and deletions shift the indices, and the correct generalisation replaces I by the set of positions whose window survives as a contiguous block of the attacked text; the identity then holds only approximately. The mean-field hypothesis on γ_w is false in real text, where entropy varies strongly with content, so that (5.3) is an *upper* bound on what survives in the wild. And nothing here concerns any undisclosed production scheme: the deployment recalled in the introduction motivates the question and is not an object of measurement.

6 Experiments

Three experiments are reported. The first verifies numerically that the propositions describe the objects the pipeline computes. The second tests Theorem 5.1 and Proposition 5.3 against exact detectors. The third applies the geometry to real chains of round-trip machine translation, and executes a test whose criterion was fixed in writing before any measurement was taken. All code, run directories and manifests are described in Section 6.4.

detours	$\delta_{\mathcal{U}}$	$\eta(\mathcal{U})$	$\text{sign}(I_n - R_{\mathcal{U}}^*)$
0	0.12241744	0.0000	(2, 0, 62)
1	0.12241744	0.2454	(2, 0, 62)
2	0.12241744	0.5190	(4, 0, 60)
3	0.12241744	0.6898	(4, 0, 60)
4	0.12241744	0.8260	(6, 0, 58)
5	0.12241744	0.9427	(6, 0, 58)
6	0.12241744	1.0465	(8, 0, 56)

Table 2: *This table summarizes the independence of the endpoint and path data, in dimension $n = 64$. All seven chains share the same first and last state, so the semantic deficit is constant to the last recorded digit; the holonomy energy is not. Note also that the signature, in the last column, moves in steps and is constant on pairs of rows, exactly as Proposition 3.1 predicts it must.*

6.1 The geometry

Over 300 pseudo-random chains in each of the dimensions $n \in \{8, 16, 64, 384\}$, with chain lengths in $\{2, 3, 4, 6, 8\}$, the worst deviations observed were: minimum eigenvalue of $I_n - R_{\mathcal{U}}^*$ equal to -6.1×10^{-15} ; negative index $n_- = 0$ in every case; $|n_+ - 2p| = 0$ in every case; $\|H\hat{v}_0 - \hat{v}_0\| = 3.3 \times 10^{-12}$; and $\|R_{\text{dir}}H - R_{\mathcal{U}}\| = 5.9 \times 10^{-12}$. This is Propositions 3.1 and 4.2 at machine precision.

Independence, part (iv) of Proposition 4.2, is exhibited directly. A family of chains with a prescribed common endpoint, wandering through an increasing number of extra dimensions, gives a semantic deficit constant at 0.12241744 with spread exactly zero, while the holonomy energy runs from 0 to 1.0465 (Table 2).

Corollary 4.3 is verified on random geodesic triangles of the two-sphere: for the 159 of 200 triangles of area at most π , the holonomy angle and the spherical excess agree to better than 1.8×10^{-9} ; the remainder agree after reduction to the principal branch, as the corollary says they must. Finally, the blindness of the signature is visible in the data: among random chains sharing the signature (8, 0, 56), the rotation energy ranges over $[0.937, 1.464]$ and the holonomy energy over $[0.365, 0.791]$.

6.2 The intact-window law against exact detectors

To test Theorem 5.1 one needs a detector whose key one holds and a generator whose entropy one controls. We therefore emit from a Zipf distribution over a vocabulary of 4000 with a freshly permuted support at each step, which fixes the entropy by construction, and watermark it with our own implementations of the green-list scheme at $h \in \{0, 1, 2, 3\}$ and of the context-seeded exponential scheme, using $\gamma = 0.25$ and $\delta_b = 2.0$. Sequences are 400 tokens long and each condition is averaged over 120 of them.

Under independent edits, the mean absolute deviation between the observed ratio and ρ^{h+1} , over the seven values of ρ from 1 down to 0.5, is 0.0023, 0.0023, 0.0022 and 0.0021 for $h = 0, 1, 2, 3$, and 0.0011 for the exponential scheme. At $\rho = 0.5$ the observed ratios are 0.499, 0.247, 0.125 and 0.065 against the predicted 0.500, 0.250, 0.125 and 0.063. Corollary 5.2 is confirmed across two decades of the ratio and across both families.

The consequence of Proposition 5.3 is more striking, and is collected in Table 3. At one and the same retention rate, the residual signal depends only on where the edits

ρ	independent	contiguous block	periodic
0.95	0.904	0.947	0.903
0.90	0.806	0.897	0.802
0.80	0.637	0.795	0.595
0.70	0.492	0.697	0.399
0.60	0.362	0.594	0.196
0.50	0.247	0.488	-0.005

Table 3: *Residual detector statistic, as a fraction of the original, for the green-list scheme with $h = 1$ under three edit patterns of identical retention rate. Counting the intact-window set directly from each edit pattern, Theorem 5.1 predicts 0.947, 0.897, 0.797, 0.697, 0.597, 0.496 for the middle column and 0.902, 0.802, 0.602, 0.401, 0.201, 0.000 for the right-hand one. The largest discrepancy is 0.008 and the typical one 0.003, over 120 sequences per cell; and the periodic pattern at $\rho = 0.5$ is predicted to give exactly zero, and does.*

fall. At $\rho = 0.5$ with $h = 1$, the same half of the tokens survives in all three columns and the residual statistic is one half of the original, one quarter of it, or nothing at all.

6.3 Real transformation chains

6.4 Reproducibility

All computations are performed by five scripts under 2. `Scripts/Article_LLW/`, documented in a local `README.md`, and every run writes a fresh timestamped directory under 7. `Results/Article_LLW/` containing its outputs, its log and a manifest recording the command, the environment, the seeds, the model revisions, the hashes of the outputs, the numerical conclusions and the known limitations. No GPU is required and none was used. The watermark keys are ours, so every detection statistic reported here is ground truth and not an estimate.

7 Conclusions and Future Developments

In this work we have taken up the formalism of linguistic loops and carried it to a place where it can be tested. We proved that the Sylvester signature proposed as the invariant of a loop is always positive semidefinite and therefore degenerates to a rank, the complete invariant being the angle spectrum; that the loop rotation is parallel transport on the unit sphere of the embedding space, so that the chain of reformulations is a path and its residue a holonomy; that this holonomy lies in the stabiliser of the initial state and is exactly what the semantic deficit discards, the two data being functionally independent; and, on the side of the detector, that the residual statistic of a context-seeded watermark is proportional to the number of positions whose seeding window survived intact.

It is mesmerizing, and we cannot help but observe it, how neatly the two halves of the picture fit. What a meaning-preserving chain preserves is an invariant; what a watermark occupies is the complement of that invariant; and the geometric object which measures the complement — a holonomy in $\text{SO}(n - 1)$ — turns out on the two-sphere to be nothing more exotic than the area swept by the path. That the same construction which was proposed as a way toward pre-verbal thought should also measure the erosion of a provenance mark is, in our humble opinion, a point in favour of the construction.

Three limitations are ours to state. The intact-window identity assumes substitutions that preserve length and a mean-field hypothesis on the strength of the mark; in real text, where entropy varies with content, it is an upper bound. The holonomy energy η is a lossy scalar reduction of H , and a null result for the former is not a null result for the latter. And the empirical layer of this article is bounded by the compute available to it: a small generator, one attack family, one embedder.

It would be then definitely interesting to analyse whether a *holonomy-aware* watermark can be constructed — one whose seeding depends on a quantity invariant under the transport, rather than on the raw token window, and which would therefore be insensitive to the arrangement of the edits in the way that Proposition 5.3 shows the present schemes are not. The semantic schemes of [6] are a first step in that direction, and it would be natural to evaluate them with the instrument developed here rather than with an endpoint similarity. In a forthcoming work we intend to study the holonomy of chains produced by learned paraphrasers rather than by translation, where the path is chosen adversarially and the geometry should be correspondingly richer. All in all, we have presented evidence that the robustness of a watermark is a functional of the path and not of its endpoint, and that the natural language in which to say so is that of parallel transport.

Use of Generative AI

The implementation of the reproducibility certificates listed in Section 6.4 was supported by AI-assisted code generation (Claude Opus, Anthropic) under the author’s direction; all certificates were independently inspected, run, and validated by the author. AI tools were also used for language editing during manuscript preparation. The author conceived the mathematical content, designed and verified the proofs, and takes full responsibility for the content of this article.

Acknowledgments

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